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Methanolysis of Polyethylene Terephthalate (PET) Catalyzed by Ionic Liquid $[Bmim]BF_4$

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ABSTRACT

The chemical recycling of polyethylene terephthalate (PET) faces challenges such as low depolymerization efficiency, harsh reaction conditions, and low product selectivity. In this study, 1-butyl-3-methylimidazolium tetrafluoroborate ($[Bmim]BF_4$) ionic liquid (IL) was developed as a catalyst for methanolysis, achieving the efficient conversion of PET into the target monomer dimethyl terephthalate (DMT). The effects of temperature, time, IL amount, and methanol quantity on the reaction were systematically investigated. The degree of depolymerization and catalyst stability were systematically evaluated by multiscale characterization (spectroscopy, chromatography, thermal analysis, etc.). Experimental results indicated that when the PET/$[Bmim]BF_4$ ratio was 6:1 and the PET/methanol ratio was 1:6 (mass ratio), reaction at 200°C for 2 h, PET was completely converted (100%), and the yield of DMT reached 82.26%. $[Bmim]BF_4$ could be reused more than five times without significant decrease in PET conversion. The kinetic results showed that the reaction was in line with the first-order kinetic model, and the apparent activation energy was about 43.24 kJ/mol. DFT calculations suggested that $[Bmim]BF_4$ efficiently catalyzed the reaction by lowering the energy barrier of the key transition state. This study provides a solution with both high efficiency and industrial potential for the green chemical recycling of PET.

1 | Introduction

Polyethylene terephthalate, usually abbreviated as PET or PETE, is a semicrystalline, thermoplastic polymer known for its high strength [1, 2]. In industry, PET is synthesized by polycondensation between ethylene glycol (EG) and terephthalic acid (TPA) or by transesterification between dimethyl terephthalate (DMT) and EG [3]. At present, plastic has been widely used in construction, medical, electronic components, automotive, agricultural packaging, and other industries [4]. So far, a variety of polymers including PET, polypropylene (PP), polyethylene (PE), polystyrene (PS), and polyvinyl chloride (PVC) have been used to produce commercial plastics [5]. In the 1960s, the consumption of plastics was only 15 million tons, and the total plastic inventory in 2020 was close to 3.2 Gt. It is estimated that it may increase to 7.7 Gt in 2050 and nearly 15 Gt in 2100 [6]. As of 2015, only about 9% of plastic waste is recycled, 12% is incinerated, and 79% is discarded in landfills, oceans, or nature [7]. At present, even in developed countries, less than 20% of postconsumer plastic waste is recycled [8]. The discarded plastic waste not only leads to the pollution of water sources, the destruction of arable land, and even penetrates into the food web. Studies have reported that various microplastics have been detected in human feces [9]. Therefore, there is an urgent need to find more efficient and green strategies to recycle this plastic waste.

An effective option to reduce plastic waste is to recycle and upgrade these products. At present, mechanical recycling and chemical recycling are the most widely used [10]. The quality of plastic recovered by mechanical methods is poor, and only low-value applications can be achieved [11]. Chemical recycling can depolymerize and reduce waste plastics into monomers, and then polymerize them into raw materials. It is still a more ideal recycling method to retain the properties of materials or the economic value of polymers. At present, the methods of degrading waste plastics mainly include alcoholysis, glycolysis, hydrolysis, and ammonolysis [12–14]. As the main raw material of plastic bottles and polyester, the recycling process of PET depolymerization into monomers mainly includes methanolysis to recover DMT [15, 16]. Recovery of bis (2-hydroxyethyl) terephthalate (BHET) by EG hydrolysis [17–19]. Hydrolysis and recovery of terephthalic acid (TPA) [20, 21]. Methanol as a solvent to depolymerize PET can reduce production costs and improve economic efficiency under the premise of ensuring the depolymerization effect. The main product DMT is an important monomer for PET synthesis. It can be used for the reproduction of PET and can realize the recycling of PET. The product has high selectivity and relatively few side reactions, which is beneficial to improve the purity and quality of the product. The solvent is easy to recycle and is environmentally friendly.

Ionic liquid (IL) is a room temperature liquid organic salt composed of positive and negative ions. It has the characteristics of nonvolatility, no pollution to the atmosphere, and stable chemical properties. It is a research hotspot of green chemistry and has broad application prospects [22–25]. In a large number of studies on the chemical recovery of PET, the main goal is to reduce the reaction time or improve the operating conditions while increasing the monomer yield [26]. Jiang et al. [27] synthesized poly (ionic liquids) (PILs) from 1-vinyl-3-ethylimidazole acetate ([VEIm]Ac) and metal acrylates to study the methanolysis of PET. The effects of reaction temperature and reaction time on the methanolysis of PET were investigated. Ma et al. [28] synthesized the IL $[HO_3S(CH_2)_3\text{-}NEt_3]Cl\text{-}[ZnCl_2]$ from Brönsted-Lewis diacid, and the alcoholysis of PET was efficiently catalyzed at 195°C. The yield of DMT was about 78.4%. These studies offer new methods for designing efficient, stable, and recyclable catalysts for PET methanolysis. However, synthesizing these ILs encounters challenges due to complex processes, challenging purification, and unstable performance, making large-scale industrial applications challenging.

This study proposes an innovative solution to these critical issues by utilizing the commercially available IL $[Bmim]BF_4$ to develop a catalytic system. Its advantages include a simple structure, excellent thermal stability, and a lower cost compared to custom ILs. Through systematic investigation of reaction kinetics and mechanisms, the scientific basis for achieving complete PET conversion and high DMT selectivity under mild conditions using this system has been clarified. This work provides a new strategy with both a theoretical foundation and technical feasibility for the industrial application of PET chemical recycling, offering significant implications for mitigating environmental pollution and energy consumption caused by waste PET.

2 | Experimental

2.1 | Materials and Instruments

PET waste bottles were recycled, washed with deionized water, and dried, then cut into 3 × 3 mm flakes for use. Anhydrous methanol (≥ 99.8% purity) was purchased from Tianjin Kemio Chemical Reagent Co. Ltd. 1-butyl-3-methylimidazolium tetrafluoroborate ($[Bmim]BF_4$, 99.0% purity) was obtained from Shanghai Chengjie Chemical Co. Ltd. (China). Standard samples, including DMT (99%), 2-hydroxyethyl methyl terephthalate (> 97%), and bis(2-hydroxyethyl) terephthalate (> 85%), were purchased from Aladdin. The following analytical instruments were employed: Fourier-transform infrared (FTIR) spectrometer (Thermo Fisher Scientific, USA); 600 MHz nuclear magnetic resonance (NMR) spectrometer (Bruker Scientific Instruments); Differential scanning calorimeter (DSC, NETZSCH); Scanning electron microscope (SEM, Hitachi High-Tech); High-performance liquid chromatograph (HPLC, Agilent, USA); Thermogravimetric analyzer (TGA, TPA Instruments, USA). The experimental device was designed and built by our laboratory (Figure S1), which consisted of a reactor with an effective volume of 50 mL and a maximum load pressure of 25 MPa.

2.2 | Characterization and Methods

The functional groups and their positions in the products were analyzed using a FT-IR (iS50) with a spectral range of 7800–100 cm^{-1}. Molecular structures were characterized by ^1H NMR (AVANCE NEO 600 M), with tetramethylsilane (TMS) as the internal standard and deuterated methanol (CD_3OD) as the solvent. For liquid chromatography analysis, the products were dissolved in a methanol–water mixture (7:3, v/v) to prepare a 40 mg/L solution. The analysis was performed under the following conditions: a C8 column (4.6 × 150 mm), SPD-1 ultraviolet detector at 254 nm, a mobile phase of methanol/water (7:3), an injection volume of 5 μL, and column temperature maintained at 25°C. Thermal properties were evaluated using DSC (DSC300) with a heating rate of 10°C/min under nitrogen flow. TGA was conducted on Q600 analyzer: samples (~5 mg) were heated from room temperature to 600°C at 10°C/min under a nitrogen atmosphere. Microstructure morphology was examined by SEM (FlexSEM1000) at an accelerating voltage of 10.0 kV, working distance of 7.7 mm, and magnification of 1.50 k, using secondary electron (SE) imaging. The microstructure was further investigated using a SC-40M optical microscope.

2.3 | Methanolysis of PET

Figure 1 illustrates the catalytic methanolysis of PET catalyzed by $[Bmim]BF_4$. The transesterification of methanol with PET produces DMT and EG.

The washed and dried PET bottle flakes and methanol were added to the reaction kettle at a mass ratio of 1:4–8, followed by the addition of [Bmim]BF_4 as catalyst (mass ratio of [Bmim]BF_4 to PET = 1:4–8). The sealed reaction kettle was placed in a heated oil bath with magnetic stirring and a thermocouple. After heating to the target temperature (170°C–220°C), the reaction proceeded for 0.5–3.0 h, and the pressure was recorded at intervals. Upon completion, the reactor was cooled to room temperature. The product was filtered to obtain solid and liquid phases. The solid-phase products were dried in an oven at 50°C to constant weight before being subjected to qualitative and quantitative analysis. The liquid-phase products, including EG, IL, and remaining methanol, were separated and recovered by rotary evaporation for recycling. The schematic diagram of the specific experimental process is shown in Figure 2.

The conversion of PET and the yield of monomer can be defined by Equations (1) and (2), respectively:

$$\text{PET conversion}(\%) = \frac{m_I - m_F}{m_I} \times 100\% \tag{1}$$

$$\text{yield}(\%) = \frac{m_A}{m_T} \times 100\% \tag{2}$$

In Equation (1), m_I represents the initial mass of PET, and m_F represents the final mass of PET after the reaction. In Equation (2), m_A represents the actual mass of monomer after the reaction, and m_T represents the theoretical mass of monomer generated by the chemical reaction of a certain mass of PET and methanol.

3 | Result and Discussion

3.1 | Characterization and Analysis of the Product

FT-IR was performed to analyze the degradation products. Figure 3 presents the infrared spectra of products obtained at various reaction times (200°C) compared with standard compounds. The spectra of the products show clear characteristic peaks at multiple wave numbers, which correspond to specific chemical bond vibrations. The strong absorption peak near 1716 cm^{-1} can be attributed to the stretching vibration of the ester group (C═O), indicating the presence of the ester group structure in the sample. The absorption peak near 1504 cm^{-1} is the skeleton vibration peak of the benzene ring. The C—H stretching vibration peak appears near 2959 cm^{-1}, corresponding to the C—H stretching vibration of the methyl group. There is a bending vibration peak of the methyl group near 1386 cm^{-1}, which can help to confirm the existence of the methyl group. There is almost no difference between the products spectrum and the standard spectrum, which indicates that the main substance in the solid-phase products is DMT, and it is relatively pure.

The ^1H NMR spectroscopy of the degradation products was carried out on a 600 MHz spectrometer (Figure 4a). The signals at δ 8.11 ppm (aromatic protons) and δ 3.94 ppm (methyl protons) exhibited an integration ratio of 5:8, closely matching the 2:3 ratio observed for the DMT standard, thus confirming DMT as the main product. The residual solvent peaks of CD_3OD (δ 4.86 ppm for H_2O, δ 3.31 ppm for CD_3OD) coincided with those in the standard spectrum (Figure S3), and no significant

FIGURE 1 | PET methanolysis catalyzed by [Bmim]BF_4.

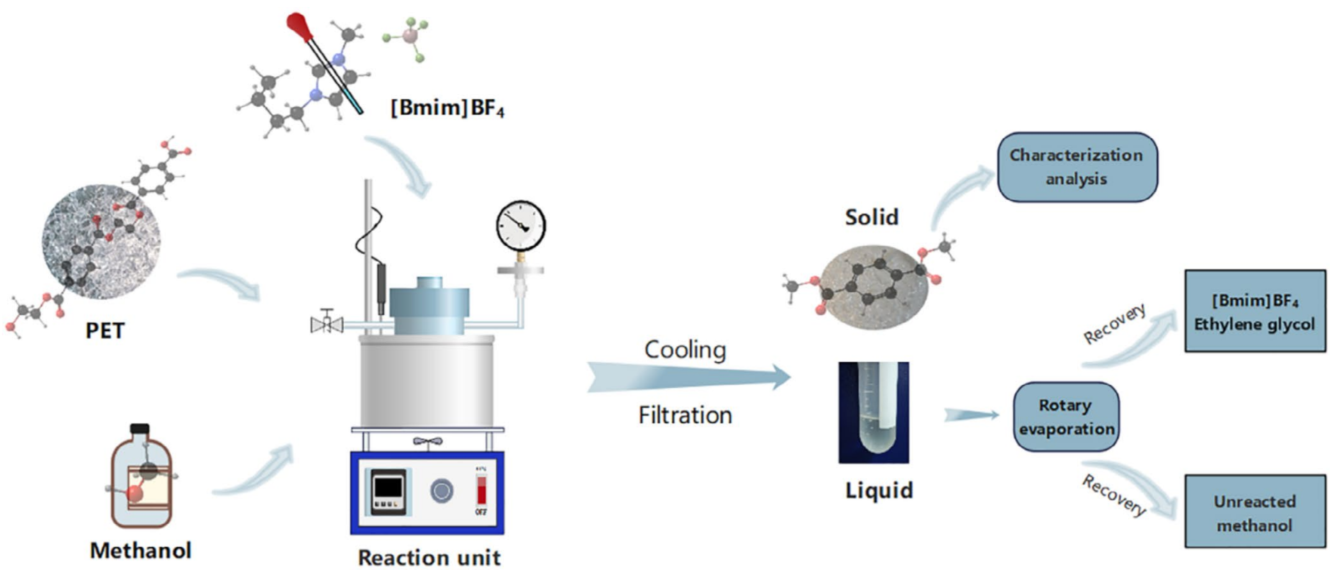

FIGURE 2 | Experimental process of PET methanol depolymerization catalyzed by [Bmim]BF_4.

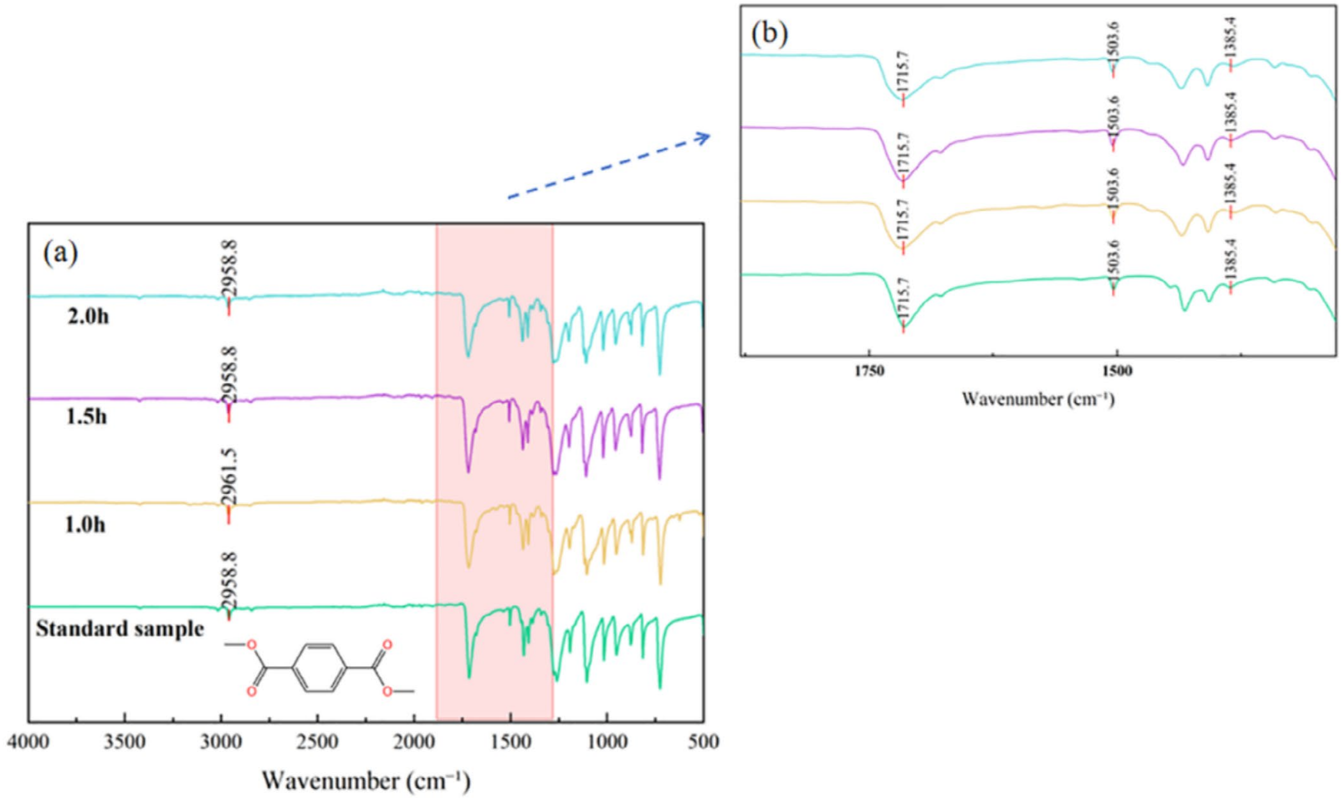

FIGURE 3 | Infrared spectra of the products and standard samples at different reaction times (200°C, m(PET):m([Bmim]BF_4) = 6:1, m(PET):m(methanol) = 1:6).

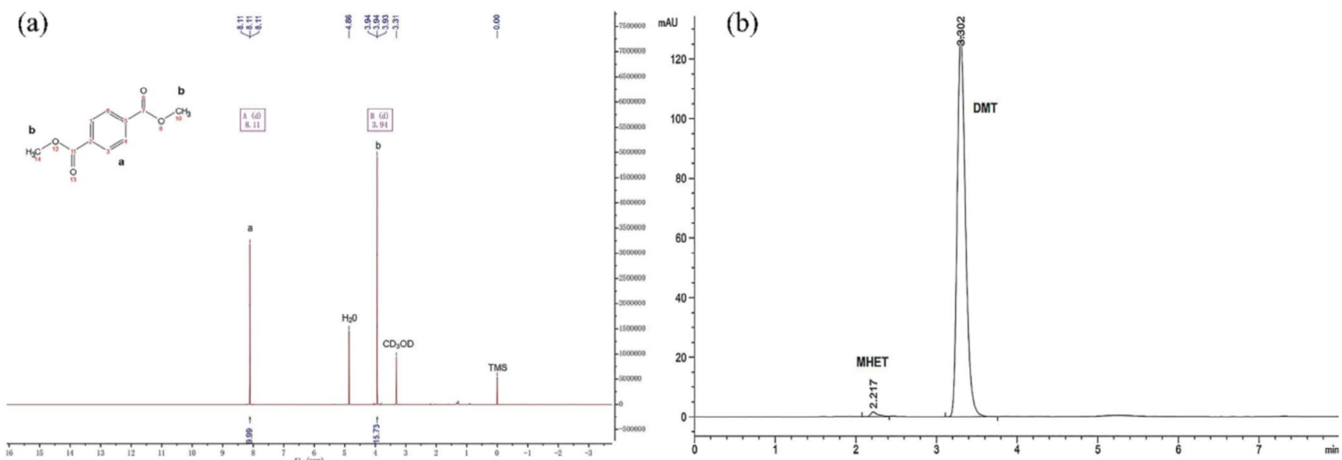

FIGURE 4 | (a) ^{1}H NMR characterization results of the degradation products. (b) Liquid chromatography characterization results (reaction conditions: 200°C, 2h, mass ratio of PET to [Bmim]BF_4 = 6:1, mass ratio of PET to methanol = 1:6).

impurity signals were detected, indicating high product purity. High-performance liquid chromatography (HPLC) analysis (Figure 4b), when compared with standard chromatograms (Figures S4 and S5), further verified the presence of DMT along with trace amounts of MHET, indicating high reaction selectivity and minimal byproduct formation.

The DSC analysis (Figure 5) revealed a main endothermic peak at 143.4°C for the degradation products, slightly lower than the 146.3°C observed for the DMT standard. The difference in the melting peak temperature suggests that the product purity is high but may contain trace amounts of intermediates. Overall, the degradation products exhibit similar melting characteristics to DMT.

SEM and optical microscopy observations (Figure 6) revealed that the pristine PET surface was smooth with uniformly aligned molecular chains (Figure 6a,c). After degradation, the material displayed blocky/fractured crystalline structures with well-defined boundaries (Figure 6b,d). Macroscopically, PET fragments transformed from transparent (Figure 6e) to white crystalline solids (Figure 6f), confirming that ionic-liquid catalysis achieved PET depolymerization and generated a new microstructural morphology.

TGA and DSC of the [Bmim]BF_4 were conducted (Figure 7). TGA results showed that the mass remained stable between 20°C and 200°C, with no significant thermal decomposition. A slight mass loss was observed between 200°C and 300°C, indicating initial thermal degradation. The main decomposition phase began around 300°C, with the maximum decomposition rate occurring at 450°C, and complete decomposition nearing 470°C. DSC analysis further revealed a stable heat flow curve up to 200°C, indicating structural stability. A sharp decline was observed starting at 300°C, with a significant endothermic peak forming at 455°C, closely matching the temperature of the maximum decomposition rate in TGA. No notable exothermic or endothermic events were observed after 500°C. The combined TGA and DSC analyses demonstrated that [Bmim]BF_4 is suitable for applications at moderate to low temperatures, with significant decomposition occurring above 300°C, providing essential thermodynamic data for its use in high-temperature applications.

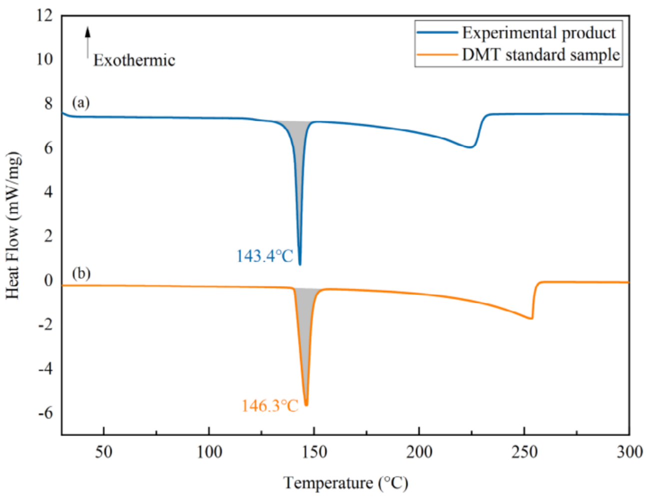

FIGURE 5 | Comparison of differential scanning calorimetry (DSC) thermograms of the degradation products and the DMT standard.

3.2 | Methanolysis of PET Catalyzed by [Bmim]BF_4

To determine the optimum conditions for PET methanolysis in the [Bmim]BF_4 catalytic system, we systematically investigated four key parameters: reaction temperature, reaction time, catalyst dosage, and methanol dosage.

Figure 8a illustrates the temperature effect on PET conversion and DMT yield. Below 170°C, no reaction occurred. The conversion and yield increased with temperature, reaching complete PET conversion (100%) and maximum DMT yield (82.26%) at 200°C. Beyond this temperature, the DMT yield declined, which might be because when the temperature exceeds 200°C, ILs will decompose slightly, resulting in a reduction of catalytic active sites. That is not conducive to improving the selectivity of the product DMT. Figure 8b shows the effect of reaction time on PET conversion and DMT yield. The PET conversion is only 36% at 0.5 h, but increased rapidly thereafter. At 1.0 h and above, the PET conversion remains constant at 100%. The DMT yield was initially low at 0.5 h but increased with the increase of time, peaking at 2.0 h before declining. This indicate that the relatively short reaction time results in incomplete PET degradation, whereas prolonged reaction promotes monomer oligomerization and higher energy consumption [29]. Figure 8c shows that both PET conversion (< 10%) and DMT yield (≈0%) were negligible without [Bmim]BF_4.

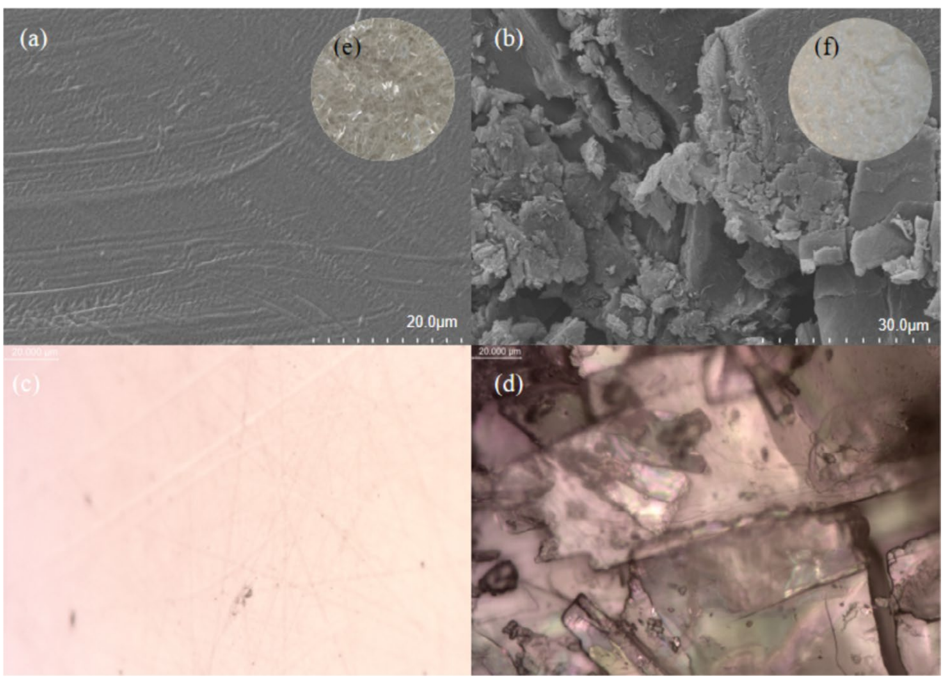

FIGURE 6 | SEM and optical microscopic images of PET before and after degradation.

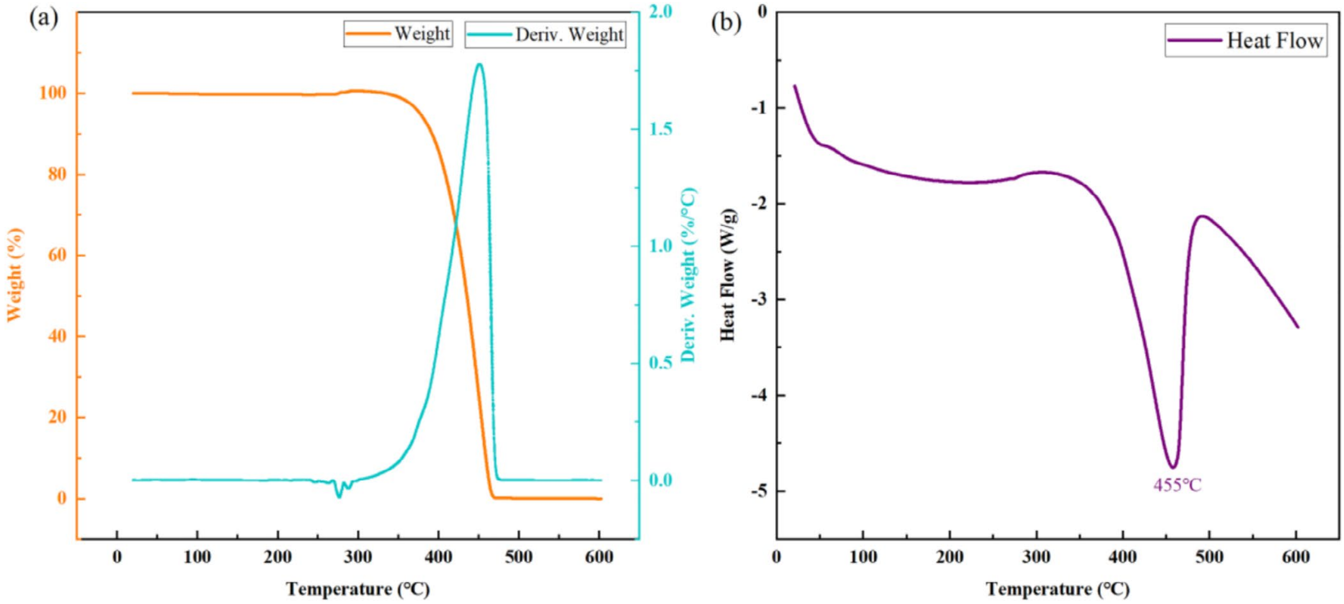

FIGURE 7 | Thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) results of $[Bmim]BF_4$.

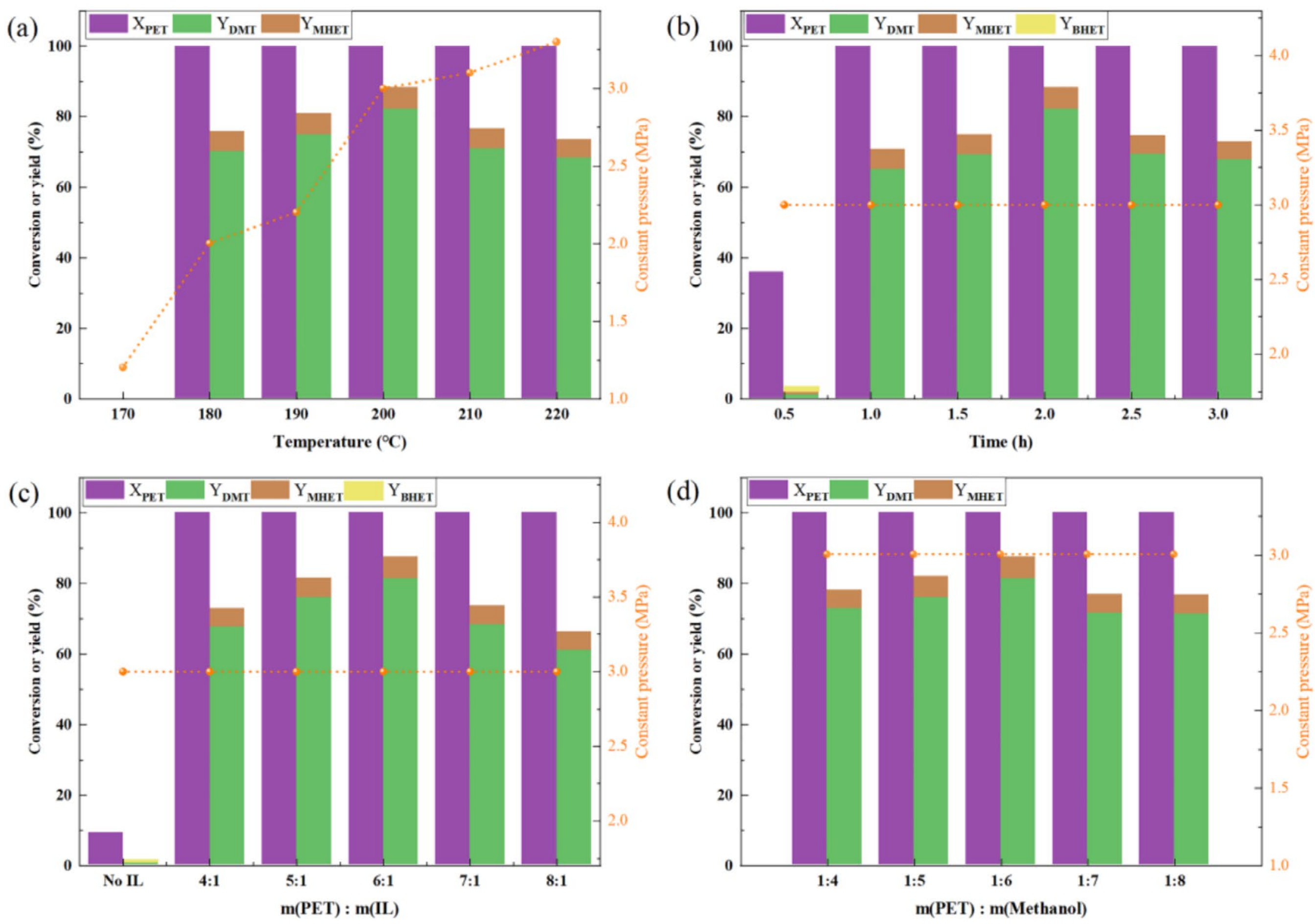

FIGURE 8 | The effects of different factors on the methanolysis of PET catalyzed by $[Bmim]BF_4$: (a) reaction temperature; (b) reaction time; (c) catalyst dosage; (d) methanol dosage. If not specified, the default reaction conditions are as follows: Reaction temperature (200°C), reaction time (2 h), m (PET):m (IL) = 6:1, m (PET):m (CH_3OH) = 1:6.

When the m (PET):m (IL) ratio reached 6:1, the DMT yield peaked, confirming that the IL is crucial for PET depolymerization. However, excessive IL increased the system viscosity, impairing reactant diffusion and DMT formation, whereas insufficient amounts led to poor catalytic performance. The methanol dosage was also investigated (Figure 8d). The optimal m(PET):m(CH_3OH) ratio was 1:6. Further increasing methanol content reduced the reaction efficiency, likely because excess methanol diluted the [Bmim]BF_4 concentration, thereby weakening its catalytic effect [30]. Notably, only trace amounts of MHET and BHET intermediates were detected, demonstrating the high selectivity of [Bmim]BF_4 toward DMT generation. In summary, the ideal conditions for the reaction are as follows: the reaction temperature is 200°C, the reaction time is 2 h, m (PET):m (IL) = 6:1, m (PET):m (CH_3OH) = 1:6.

The PET conversion remained at 96% after five reuse cycles of [Bmim]BF_4 (Figure 9), indicating outstanding cycling stability of the IL under optimal conditions.

3.3 | Kinetics of PET Catalyzed by [Bmim]BF_4

Methanol depolymerization of PET is usually considered a first-order reaction [27, 28] Methanol is always excessive during the reaction. The reaction rate of PET in methanol can be expressed by the following equation:

$$r = \frac{dC_{\mathrm{PET}}}{dt} = -kC_{\mathrm{PET}} \tag{3}$$

where k is the rate constant of the reaction, and C_{PET} represents the concentration of PET at time t. Since PET is a solid, the mass is used to replace the concentration, and the reaction rate equation can be transformed into following equation.

$$r = -\frac{dm_{\mathrm{PET}}}{dt} = \frac{dm_{\mathrm{DMT}}}{dt} = kf\left(m_A/m_T\right) \tag{4}$$

The DMT yield can be defined as:

$$Y = \frac{m_A}{m_T} \tag{5}$$

where m_A represents the actual mass of DMT after the reaction t, and m_T represents the theoretical mass of DMT generated by the chemical reaction of a certain mass of PET and methanol. Suppose that $f\left(m_A/m_T\right) = \left(m_A/m_T\right)^n$, n denotes the reaction order, then $f(Y) = Y^n$, Equation (5) can be written as follows:

$$\frac{d_Y}{d_t} = kY^n \tag{6}$$

when $n = 1$, through the integral Equation (6), we can get:

$$lnY = kt \tag{7}$$

According to the Arrhenius equation:

$$lnk = -\frac{E_a}{\mathrm{RT}} + \ln\mathrm{A} \tag{8}$$

The reaction activation energy E_a can be obtained by Equation (8), where A is the Arrhenius constant, R is the gas constant, and T is the Kelvin temperature. With $1/T$ as the abscissa and ln k as the ordinate, Figure 10 is drawn as follows:

The Arrhenius plot analysis yielded an activation energy of 43.24 kJ/mol and the pre-exponential factor A is 229.75. Thus, the kinetic equation for [Bmim]BF_4 catalyzed PET methanolysis can be expressed as:

$$lnk = -\frac{43.24}{\mathrm{RT}} + 5.44 \tag{9}$$

Table 1 summarizes the comparison of PET methanolysis results from literatures. The activation energy of this work is significantly lower than that in the literature, which indicates that the method used in this paper can reduce the activation energy and accelerate the reaction rate.

3.4 | Discussion on the Reaction Mechanism

Based on the results of this study, a possible reaction mechanism of PET methanol depolymerization catalyzed by [Bmim] BF_4 was proposed (Figure 11). Under the appropriate reaction

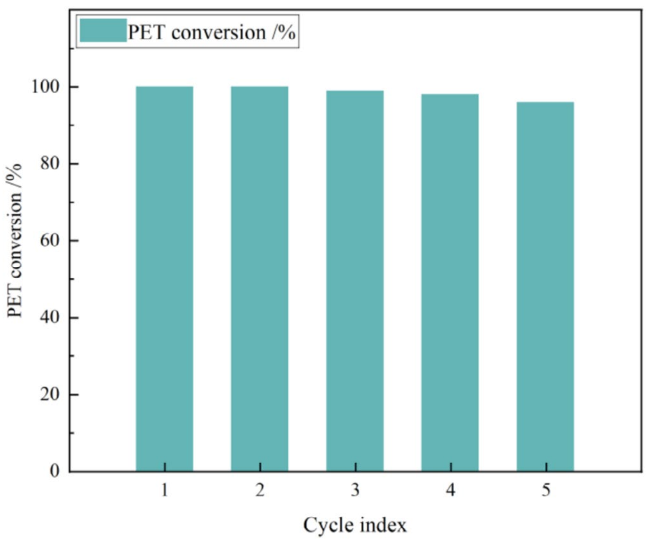

FIGURE 9 | The effect of repeated use of [Bmim]BF_4 on the degree of degradation.

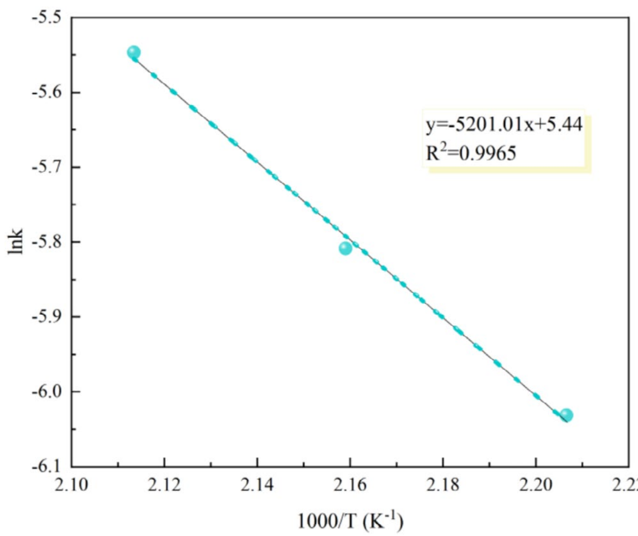

FIGURE 10 | Arrhenius plot of methanolysis of PET catalyzed by IL [Bmim]BF_4.

TABLE 1 | Comparison of PET methanolysis literature.

Catalyst	Temperature (°C)	Time (h)	DMT yield (%)	Activation energy (KJ/mol)	References
—	310	1.0	97.7	49.9	[31]
PIL-Zn^{2+}	170	1.0	89.1	107.6	[27]
DES	120	2.0	80.0	106.9	[32]
DBN/Phenol	130	1.0	95.3	103.3	[33]
K_2CO_3	25	24.0	93.1	66.5	[15]
[Bmim]BF_4	200	2.0	82.3	43.2	This work

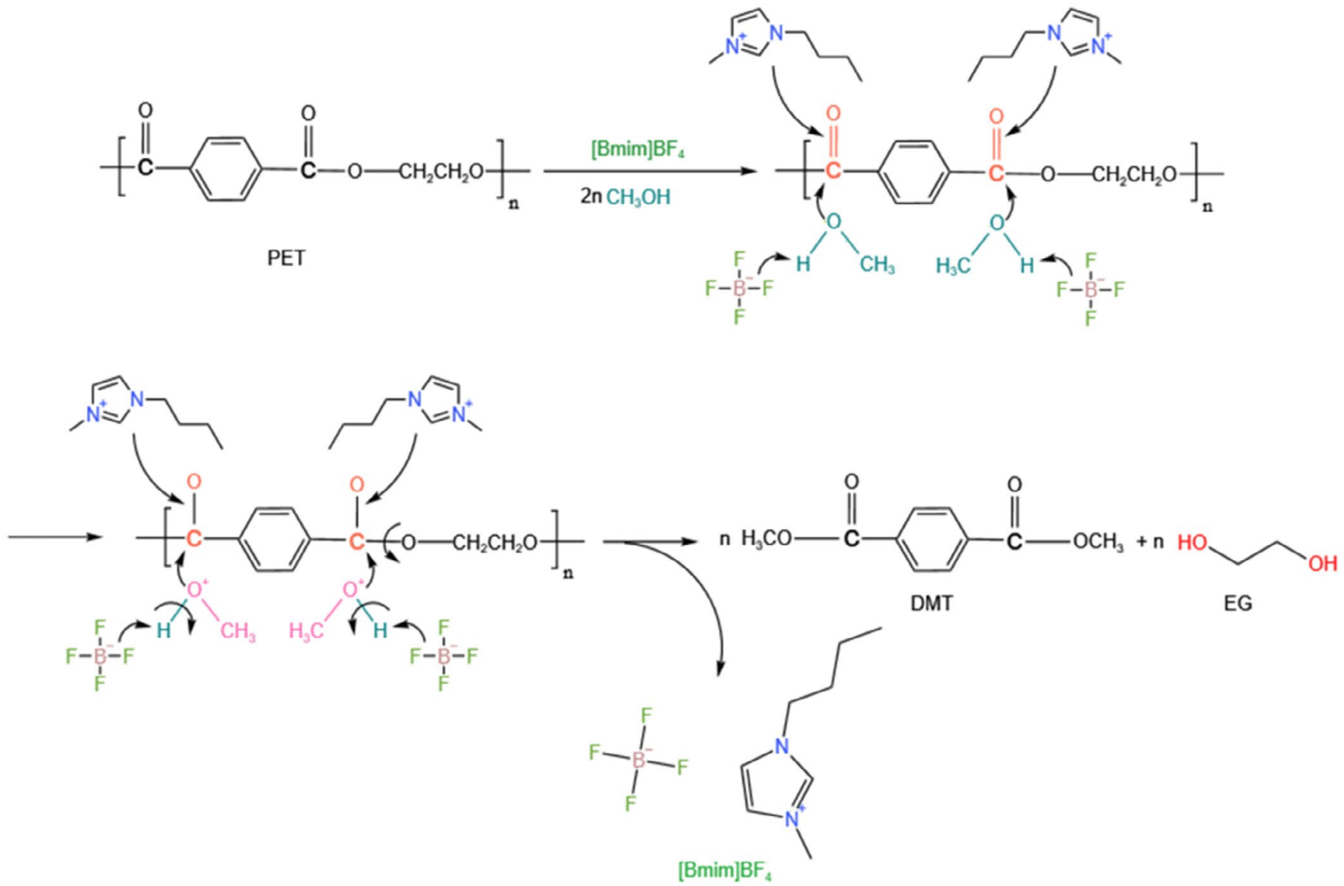

FIGURE 11 | The possible reaction mechanism of PET methanolysis catalyzed by [Bmim]BF_4.

conditions, the cation [Bmim]$^+$ of the IL polarizes the PET carbonyl group (C=O) through hydrogen bonding to enhance its electrophilicity, the anion BF_4^- activates methanol to form methoxy (CH_3O^-), which then efficiently attacks the PET carbonyl carbon in the cationic-regulated nucleophilic microenvironment and synergistically promotes the depolymerization reaction. The methoxyanion is a strong nucleophilic reagent with high reactivity toward the ester bond in the PET chain. It attacks the carbonyl carbon atom of the PET chain, where the oxygen from methanol binds to the C on the polyester C=O to form a new ester group, leading to chain breaks to form DMT and EG. DMT forms via the combination of the terephthalic acid part of the original PET molecular chain with two methoxy groups, while EG is the remaining part after the ester bond is broken. The [Bmim]BF_4 catalyst can continue to participate in the next round of reactions and play a catalytic cycle role.

DFT calculations reveal that the [Bmim]BF_4 significantly reduced the transition state energy barriers of TS1 and TS2 in PET methanol depolymerization, where TS1 is the rate-limiting step. Although the IL slightly increases the energy of the final state (FS2), the markedly reduced transition state energy barrier combined with enhanced exothermicity of FS1 collectively drives the forward reaction. Kinetic analysis demonstrates that [Bmim]BF_4 effectively accelerates the reaction by lowering the activation energy while improving system stability. Therefore, [Bmim]BF_4 serves as an efficient catalytic medium for PET methanol depolymerization, with the mechanism primarily involving transition state stabilization and kinetic regulation (Figure 12).

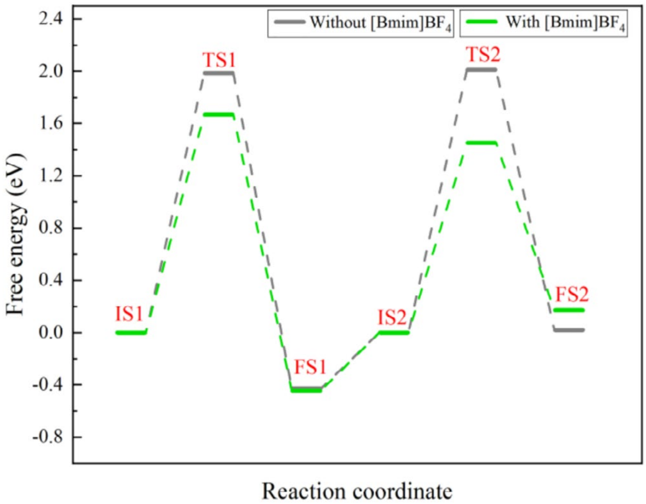

FIGURE 12 | The energy change of PET methanolysis reaction with or without $[Bmim]BF_4$.

4 | Conclusions

This study demonstrates that the IL $[Bmim]BF_4$ is an efficient and stable catalyst for PET methanolysis. Under the reaction conditions of 200°C and 2h, with PET/IL and PET/methanol mass ratios of 6:1 and 1:6, respectively, PET is completely converted (100%) with a DMT yield of 82.26%. Kinetic analysis reveals that the reaction follows first-order kinetics, with an apparent activation energy of 43.24kJ/mol. Mechanistic studies indicate a synergistic catalytic mechanism where the $[Bmim]^+$ cation activates PET's carbonyl group through hydrogen bonding while the BF_4^- anion promotes methanol dissociation. DFT calculations confirmed that $[Bmim]BF_4$ significantly lowers the transition state energy of the reaction. The IL demonstrated good stability and can be reused over five times without significant activity loss.

This study offers an environmentally friendly catalytic strategy for efficient PET waste recycling, while also providing insights into the chemical recycling of other polyesters. Future research will focus on the industrial scale-up and the development of continuous processes for this catalytic system, aiming to advance the practical application in waste plastic resource utilization.

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Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.